Sensitivity Analysis

In LP, the parameters (input data) of the model can change within certain limits without causing changes in optimum. This is referred to as sensitivity analysis.

Understanding sensitivity analysis graphically

Two cases will be considered:

1. Sensitivity of the optimum solution to changes in the availability of the resources (RHS of the constraints.
2. Sensitivity of the optimum solution to changes in the unit profit or unit cost (coefficients of the objective function)

Changes in the RHS → right hand side

Ex: JOBCO manufactures two products on two machines. A unit of product 1 requires 2 hours on machine 1 and 1 hour on machine 2. For product 2, one unit requires 1 hour on machine 1 and 3 hours on machine 2. Revenues per product 1 and 2 are $30 and $20, respectively. Total daily processing time available for each machine is 8 hours.

Sol: Let x_1 and x_2 represent daily number of unit of products 1 and 2, respectively. Then LP model is

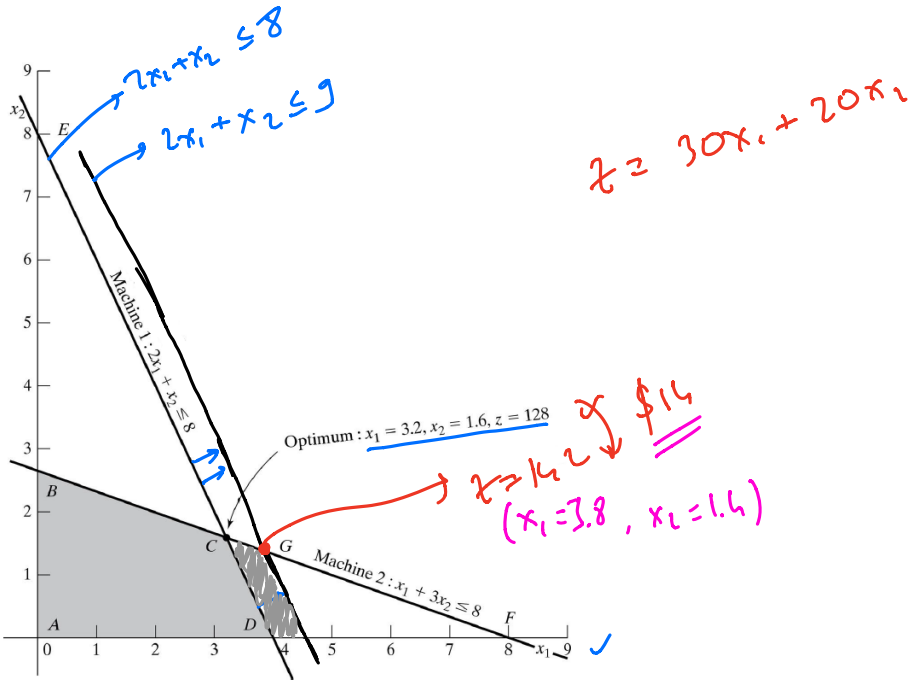
maximize $30x_1 + 20x_2$

s.t

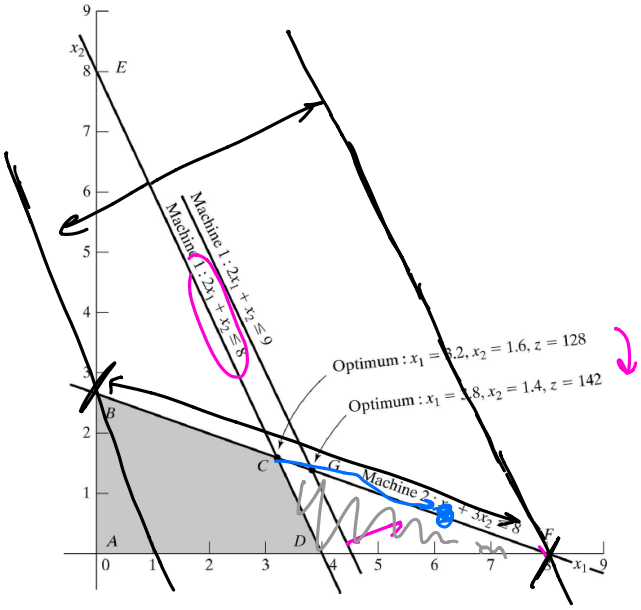
$$2x_1 + x_2 \le 8 \quad \text{(Machine 1)}$$

$$x_1 + 3x_2 \le 8 \quad \text{(Machine 2)}$$

$$x_1, x_2 \ge 0$$



- If daily capacity of M1 is increased from 8 to 9 hours, the new optimum will move to point G.



- The rate of change in optimum z resulting from changing M1 capacity can be calculated as:

$$\textit{Rate of revenue change resulting from increasing M1 capacity by 1 hour } (C \rightarrow G) = \frac{z_G - z_C}{\textit{Capacity change}} = \frac{142 - 128}{1} = \$14/\text{hr}$$

- This rate is a direct link between the model input (resources) and its output (total revenue).

8 → 13 5 × $14 = $70

- It says that a unit increase (decrease) in M1 capacity will increase (decrease) revenue by $14.

- **Unit worth of a resource** is an apt description of the rate of change of the objective function per unit change of a resource. (Note that **dual** or **shadow** price is standard in all LP literature and software packages.)

- Under which conditions is this **dual price** of M1 is valid? → It's along the BF line.

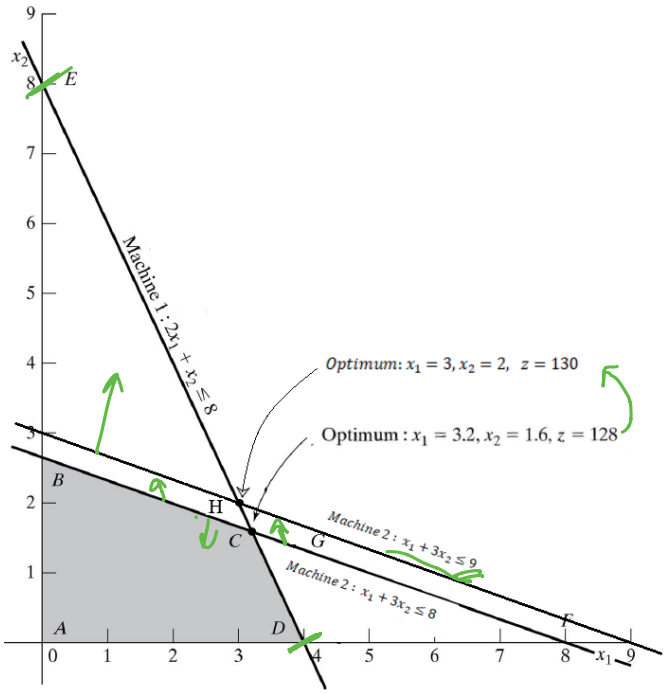
At $B(0, 2.67)$: $2x_1 + x_2 = 2\times0 + 2.67 = 2.67$ hrs.

At $F(8, 0)$: $2x_1 + x_2 = 2\times8 + 0 = 16$ hrs. [2.67, 16]

- So, $14/hr remains valid only in the range

2.67 hr ≤ M1 capacity ≤ 16 hr.

Similarly calculate dual price for M2 capacity



$x_1 = 3, x_2 = 2, z = 130$

$$Optimum: \quad \frac{Z_H - Z_C}{Capacity\ change} = \frac{130 - 128}{1} = \$2/hr$$

- Dual price = $\$2/hr$
- It is valid on **DE** line:

At $D(4,0) = x_1 + 3x_2 = 4$ hrs

At $E(0,8) = x_1 + 3x_2 = 24$ hrs.

Therefore, the dual price of \$2/hr for M2 remains applicable for the range

$4 \text{ hrs} \leq M2 \text{ capacity} \leq 24$ hrs.

The reported limits for M1 and M2 are referred to as the **feasibility ranges**.

dual price M1 → $14/hr
" M2 → $2/hr

The dual prices allow making economic decisions about the LP problem:

1) If JOBCO can increase the capacity of both machines, which should receive priority?

- obviously M1, because of $14/hr dual price vs $2/hr (M1* / M2)
- Each additional hour of M1 increases the revenue by $14.

2) A suggestion is made: Increase the capacities of M1 and M2 at the additional cost of $10/hr for each.

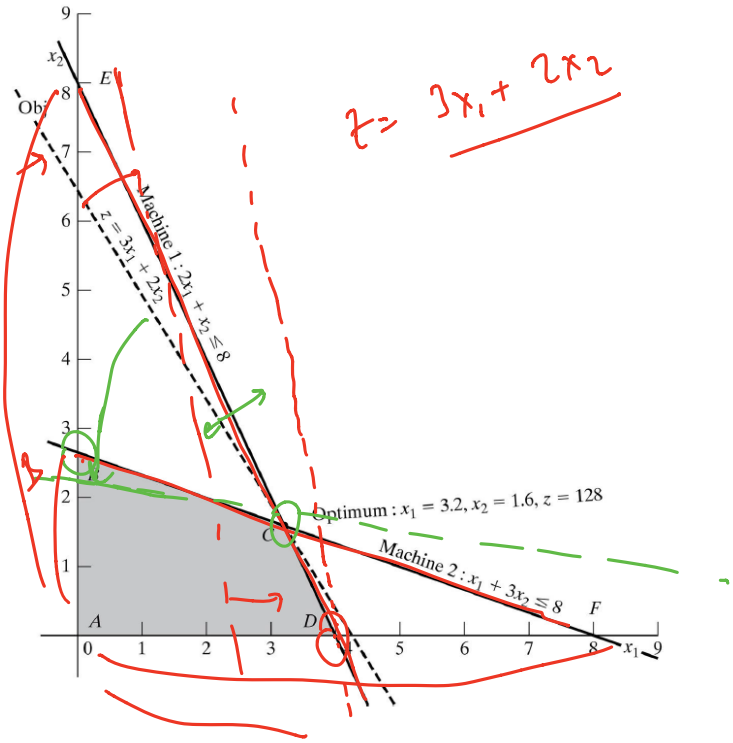
- For M1, the additional net revenue per hour is 14 − 10 = $4
- For M2: 2 − 10 = −$8
- Therefore, only M1 capacity should be increased.

3) If capacity of M1 is increased from 8 to 13 hours, how will impact the optimum revenue.

$14 \times (13-8) = 14 \times 5 = \$70$

Because, feasibility range for M1 is [2.67, 16] and 13 is in this range.

Changes in the Objective Function



- Optimum occurs at point C
- Changes in revenue units (i.e. the objective function coefficients) will change the slope of z. However, the optimum solution at point C remains unchanged as long as the objective function lies between BF and DE.

- General format of objective function: maximize $z = c_1x_1 + c_2x_2$
- The optimum solution will remain at point C as long as $z = c_1x_1 + c_2x_2$ lies between the two lines $2x_1 + x_2 = 8$ and $x_1 + 3x_2 = 8$. This means that the ratio $\frac{c_1}{c_2}$ can vary between $\frac{1}{3}$ and $\frac{2}{1}$ which yields the following **optimality range**:

$$\frac{1}{3} \leq \frac{c_1}{c_2} \leq \frac{2}{1}$$

Observe here that $\frac{c_1}{c_2}$ is the tangent of objective function, $\frac{1}{3}$ is the tangent of constraint 2 and $\frac{2}{1}$ is the tangent of constraint 1.

The following questions are answered after this analysis:

1) Suppose the unit revenues for products 1 and 2 are changed to \$35 and \$25, respectively.
   - Will the current optimum remain the same?

new objective function: $z = 35x_1 + 25x_2$ → new z: $35 \times 3.2 + 25 \times 1.6 = \$152$

Thus, $\frac{c_1}{c_2} = \frac{35}{25} = 1.4 \rightarrow \frac{1}{3} \leq 1.4 \leq 2$

So, point C remains optimum.

2) Suppose that the revenue of product 2 is fixed at its current value $c_2 = \$20$.
   - What is the associated optimality range for the unit revenue for product 1, c_1, that will keep the optimum unchanged?

$$\frac{1}{3} \leq \frac{c_1}{20} \leq \frac{2}{1} \Rightarrow \frac{20}{3} \leq c_1 \leq 40$$

$$6.67 \leq c_1 \leq 40$$

   - We can similarly determine the optimality range for c_2 by fixing the value of c_1 at \$30.

$$\frac{1}{3} \leq \frac{30}{c_2} \leq \frac{2}{1} \Rightarrow 15 \leq c_2 \leq 90$$

Algebraic Sensitivity Analysis – Changes in the RHS

In previous section, we made a sensitivity analysis using the graphical solution. This section extends the analysis to the general LP model with an example.

Ex: TOYCO uses three operations to assemble three types of toys: trains, trucks and cars.

- Daily availability of the three operations are 430, 460 and 420 minutes, respectively.
- Revenues per unit of toy train, truck and car are $3, $2 and $5, respectively.
- Assembly time per train at the three operations are 1, 3, 1 minutes, respectively.
- The corresponding times per truck and per car are (2,0,4) and (1,2,0) minutes respectively.

Sol: Let x_1, x_2, x_3, represent daily number of units assembled of trains, trucks and cars, respectively. Then the model is,

Maximize $z = 3x_1 + 2x_2 + 5x_3$

S.t

$x_1 + 2x_2 + x_3 \le 430$ (operation 1)

$3x_1 + 2x_3 \le 460$ (" 2)

$x_1 + 4x_2 \le 420$ (" 3)

$x_1, x_2, x_3 \ge 0$

opt → 0 trains, 100 trucks, 230 cars

$z = 3\times0 + 2\times100 + 5\times230 = 1350$

Using x_4, x_5, x_6 as the slack variables for constraints of operation 1, 2 and 3, respectively, the optimum tableau is:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	4	0	0	1	2	0	1350
x_2	-1/4	1	0	1/2	-1/4	0	100
x_3	3/2	0	1	0	½	0	230
x_6	2	0	0	-2	1	1	20

$x_1 = 0$, $x_4 = 0$, $x_5 = 0$ → nonbasic

Solution recommends 0 trains, 100 trucks, 230 cars, z=$1350

Suppose D_1, D_2 and D_3 are the positive (or negative) changes made in the allocated daily manufacturing time of op1, op2 and op3, respectively. Then TOYCO model can be changed to

Maximize $z = 3x_1 + 2x_2 + 5x_3$

s.t $x_1 + 2x_2 + x_3 \leq 430 + D_1$ (operation 1)

$3x_1 + 2x_3 \leq 460 + D_2$ (operation 2)

$x_1 + 4x_2 \leq 420 + D_3$ (operation 3)

$x_1, x_2, , x_3 \geq 0$

Rewrite the starting tableau using the new RHS:

starting tableau

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	D_1	D_2	D_3
z	-3	-2	-5	0	0	0	0	0	0	0
x_4	1	2	1	1	0	0	430	1	0	0
x_5	3	0	2	0	1	0	460	0	1	0
x_6	1	4	0	0	0	1	420	0	0	1

Observe that shaded regions are identical. Simplex operations yield the same table. That is,

optimal tableau

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution	D_1	D_2	D_3
z	4	0	0	1	2	0	1350	1	2	0
x_2	-1/4	1	0	1/2	-1/4	0	100	1/2	-1/4	0
x_3	3/2	0	1	0	½	0	230	0	½	0
x_6	2	0	0	-2	1	1	20	-2	1	1

This optimal tableau provides the following optimal solution.

$z = 1350 + D_1 + 2D_2$

$x_2 = 100 + \frac{1}{2}D_1 - \frac{1}{4}D_2$

$x_3 = 230 + \frac{1}{2}D_2$

$x_6 = 20 - 2D_1 + D_2 + D_3$

$z = 1350 - \frac{1}{4}x_1 - x_4 - 2x_5$

This solution gives us the dual prices and feasibility ranges.

$z = 1350 + D_1 + 2D_2 + 0D_3$ show that

1. A unit change in operation 1 capacity ($D_1 = \pm 1$ *min.*) changes z by $1
2. A unit change in operation 2 capacity ($D_2 = \pm 1$ *min.*) changes z by $2
3. A unit change in operation 3 capacity ($D_3 = \pm 1$ *min.*) changes z by $0

So, dual prices are 1, 2 and 0 ($/min) for operation 1, 2 and 3, respectively.

Feasibility range: Current solution (optimal tableau) remains feasible if all the basic variables remain nonnegative (the non-negativity constraint of all variables) – that is

$$x_2 = 100 + \frac{1}{2}D_1 - \frac{1}{4}D_2 \geq 0$$

$$x_3 = 230 + \frac{1}{2}D_2 \geq 0$$

$$x_6 = 20 - 2D_1 + D_2 + D_3 \geq 0$$

Note that simultaneous changes on D_1, D_2 and D_3 that satisfy the inequalities will keep the solution feasible.

Ex: Let new capacities of operation 1, 2 and 3 are 480, 440 and 400. Then $D_1 = 50, D_2 = -20$ and $D_3 = -20$. Then,

$x_2 = 100 + \frac{1}{2}50 - \frac{1}{4}(-20) = 130 \geq 0$ (feasible)

$x_3 = 230 + \frac{1}{2}(-20) = 220 \geq 0$ (feasible)

$x_6 = 20 - 2\times 50 + (-20) + (-20) = -170 \not\geq 0$ (infeasible)

Therefore, current solution does not remain feasible.

Ex: If $D_1 = -30, D_2 = -12$ and $D_3 = 10$, then, $x_2 = 88, x_3 = 224, x_6 = 78$, all greater than 0. So all feasible and new solution is $x_2 = 88, x_3 = 224, x_6 = 78, z = 1296$.

$z = 1350 + D_1 + 2D_2$

$= 1350 - 30 - 2\times 12 = 1296$

Above given conditions can produce the individual feasibility ranges.
For example, a change in operation 1 only means that $D_2 = 0$ and $D_3 = 0$. Then,

$x_2 = 100 + \frac{1}{2}D_1 \geq 0$ ➔ $D_1 \geq -200$

$x_3 = 230 > 0$

$x_6 = 20 - 2D_1 \geq 0$ ➔ $D_1 \leq 10$

Therefore, $-200 \leq D_1 \leq 10$.

Similarly, feasibility ranges for operation 2 and operation 3 are $-20 \leq D_2 \leq 400$ and $-20 \leq D_3 \leq \infty$.

Note that dual prices remain applicable for any simultaneous change that keeps the solution feasible, even if they violate the individual ranges.
Ex: $D_1 = 30, D_2 = -12$ and $D_3 = 100$ will keep the solution feasible even though $D_1 = 30$ violates its range.

Algebraic Sensitivity Analysis – Objective Function

Reduced Cost: Remember the TOYCO example. The objective z-equation in the optimal tableau can be written as $z = 1350 - 4x_1 - x_4 - 2x_5$. The optimum solution does not produce toy trains ($x_1 = 0$). Reason is in z equation: a unit increase in x_1 decreases z by \$4.

$$(Reduced\ cost\ per\ unit) = (cost\ of\ consumed\ resources\ per\ unit) - (revenue\ per\ unit)$$

To understand better, consider revenue per unit for toy trucks (\$2) is less than toy train (\$3). But optimal solution recommends producing toy trucks (x_2=100) and no toy rains. Reason is that cost of resources used by one toy truck is smaller than its unit price. (the z-row in the optimal tableau tells the reduced costs for non-basic variables. E.g. it is -4 for x_1.)

In TOYCO model, let d_1, d_2 and d_3 represent the change in unit revenues for toy tucks, trains and cars, respectively. The objective function then becomes:

- Maximize $z = (3 + d_1)x_1 + (2 + d_2)x_2 + (5 + d_3)x_3$
- z-row in the starting tableau appears as:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	solution
z	$-3 - d_1$	$-2 - d_2$	$-5 - d_3$	0	0	0	0

- Optimal tableau at the end becomes:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	solution
z	$4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1$	0	0	$1 + \frac{1}{2}d_2$	$2 - \frac{1}{4}d_2 + \frac{1}{2}d_3$	0	$1350 + 100d_2 + 230d_3$

- The new optimal is the same as the original optimal tableau except for the reduced costs (z-equation coefficients). This means that changes in the objective function coefficient can affect the optimality of the problem only.
- Then, reduced cost for x_1 is $4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1$.
- The current solution remains optimal as long as the new reduced costs remain nonnegative (for maximization problems). (Remember, the algorithm continues until there is no negative value in z-row for maximization problems.)
- We, thus have the following simultaneous optimality conditions corresponding to x_1, x_4 and x_5.

$$4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1 \geq 0$$
$$1 + \frac{1}{2}d_2 \geq 0$$
$$2 - \frac{1}{4}d_2 + \frac{1}{2}d_3 \geq 0$$

Ex: Suppose objective function changed from $z = 3x_1 + 2x_2 + 5x_3$ to $z = 2x_1 + x_2 + 6x_3$. Then, $d_1 = -1, d_2 = -1$ and $d_3 = 1$.
Substitution yields

$$4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1 = 6.75 \geq 0$$
$$1 + \frac{1}{2}d_2 = 0.5 \geq 0$$
$$2 - \frac{1}{4}d_2 + \frac{1}{2}d_3 = 2.75 \geq 0$$

- All reduced costs remain nonnegative, therefore current solution remains optimal with a new value of $z = 1350 + 100d_2 + 230d_3 = \$1480$.

- For minimization case, the only difference is that reduced costs (z-equation coefficients) must be ≤ 0 to maintain optimality.
- The optimality ranges dealing with changing d_j one at a time can be developed from the simultaneous optimality conditions. For example if $d_1 = 0$ and $d_3 = 0$ and coefficient of x_2 is changed to $2 + d_2$, then optimality conditions reduce to:

$4 - \frac{1}{4}d_2 \geq 0$ ➔ $d_2 \leq 16$
$1 + \frac{1}{2}d_2 \geq 0$ ➔ $d_2 \geq -2$
$2 - \frac{1}{4}d_2 \geq 0$ ➔ $d_2 \leq 8$

Therefore, it turns out that $-2 \leq d_2 \leq 8$.

- $-2 \leq d_2 \leq 8$ translates to unit revenue as follows: $0 \leq$ *unit revenue* ≤ 10 because unit revenue is $2 + d_2$ for toy truck.

- Similarly individual changes $3 + d_1$ and $5 + d_3$, for x_1 and x_3 yield the optimality ranges $d_1 \leq 4$ and $d_2 \geq \frac{-8}{3}$, respectively.